ERDOS RESEARCH

Mathematical Foundations of Artificial Intelligence

A Universal Framework for Understanding Modern AI

GLOSSARY

Organised through the Five Lenses and the Frontier

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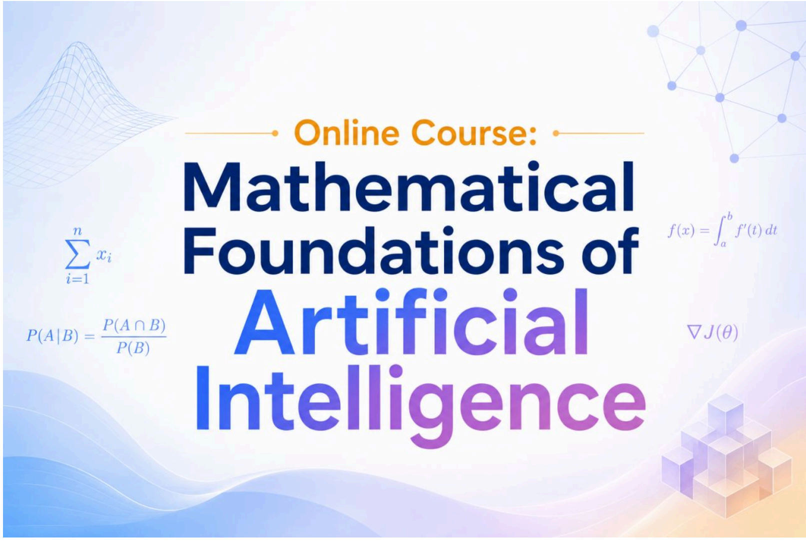
LEARNING · REPRESENTATION · COMPUTATION
OPTIMISATION · THEORY · FRONTIERS



About this glossary

How to read the entries

This Glossary relates to my course / book

[ONLINE COURSE - MATHEMATICAL FOUNDATIONS OF ARTIFICIAL INTELLIGENCE]

Please contact me if you want to join this course via linkedin (messaging or contacting)

This glossary is organised around the same six lenses through which the book/course reads modern AI. Each lens collects the vocabulary it requires, and entries are written so that the same term, when it appears under different lenses, can be approached from the angle of that lens. A reader encountering, for example, the manifold hypothesis under Representation should expect a geometric account; under Theory, a generalisation-bound-flavoured one.

Entries are concise by design. Each definition aims to give a working sense of the concept and its place in the wider framework, rather than a full mathematical treatment, which is the work of the chapters themselves.

The six lenses

Learning formulating the problem — what is being approximated, from what data, under what assumptions.

Representation how AI builds abstractions — embeddings, latent spaces, manifolds, the geometry of learned knowledge.

Computation how intelligence is implemented — the small set of architectural primitives from which modern systems are built.

Optimisation discovering useful solutions — gradient descent and its variants, loss landscapes, regularisation.

Theory why any of it works — the partial explanations now available for the empirical success of deep learning.

Frontiers where the answers are still being worked out — interpretability, scaling, world models, scientific ML, agentic AI.

Learning

Formulating the problem

Every AI system begins by defining the problem correctly. The Learning lens establishes what is being approximated, from what data, under what assumptions about how the data was generated, and under what constraints make generalisation possible.

Function Approximation. The view that every machine learning model is a parameterised function mapping inputs to outputs, and that training is the act of selecting one such function from a family. This framing unifies linear regression, decision trees, deep networks and transformers under a single mathematical object.

Hypothesis Space. The set H of all functions a given architecture can represent as its parameters vary. Architecture choice fixes H; data, loss, optimiser and initialisation then select one element from it.

Inductive Bias. The collection of assumptions a learning algorithm uses to choose between hypotheses equally consistent with the training data. Sources include architecture, optimiser, regularisation, initialisation and computational constraints; bias is not a defect but a precondition for generalisation.

Generalisation. The property that a model trained on a finite sample performs well on previously unseen inputs from the same distribution. The course distinguishes a hierarchy of meanings — in-distribution, transfer, few-shot, zero-shot, compositional — each demanding deeper structure in what has been learned.

Generalisation Gap. The difference between a model's expected loss on the data distribution and its empirical loss on the training sample. Controlling this gap is the central concern of statistical learning theory.

Supervised Learning. A paradigm in which the learner is given input–output pairs (x, y) drawn i.i.d. from an unknown joint distribution and asked to approximate the conditional mapping $x \to y$ by minimising a loss on the sample.

Unsupervised Learning. A paradigm in which only inputs are observed and the learner must discover structure — clusters, densities, latent factors — without explicit targets.

Self-Supervised Learning. A paradigm in which the supervision signal is constructed from the input itself, typically by masking or predicting parts of the data from other parts. The engine behind modern foundation models, since it scales with raw data rather than labels.

Reinforcement Learning. A paradigm in which an agent interacts with an environment, takes actions, observes consequences and receives delayed rewards. The learner must trade off exploration against exploitation and reason about long-horizon credit assignment.

I.I.D. Assumption. The assumption that training examples are independent and identically distributed draws from a fixed underlying distribution. Most classical generalisation results rest on it; its breakdown defines the distribution-shift problem.

In-Distribution / Out-of-Distribution. A distinction between test inputs drawn from the same distribution as the training data (in-distribution) and inputs that systematically differ in some respect (out-of-distribution). Out-of-distribution performance is the harder and more practically important challenge.

No Free Lunch Theorem. Wolpert's result that, averaged uniformly over all possible target functions, all learning algorithms perform identically. The theorem makes precise the necessity of inductive bias: there is no algorithm that generalises in the absence of assumptions about which targets are likely.

Empirical Risk Minimisation. The principle of choosing the hypothesis that minimises average loss on the training sample, as a proxy for the unobservable risk on the underlying distribution. The classical foundation on which modern learning algorithms are built.

Few-Shot Learning. Generalisation from a small handful of examples, often achieved by leveraging representations learned on a much larger pre-training corpus.

Zero-Shot Generalisation. Successful performance on a task for which the model has been given no training examples, typically by re-using a learned representation guided by natural-language instructions.

Compositional Generalisation. The ability to handle novel combinations of familiar elements — the most demanding form of generalisation and a continuing weakness of large statistical models.

Exploration–Exploitation Trade-off. The reinforcement-learning dilemma between selecting actions believed to be high-value (exploitation) and selecting actions whose value is uncertain (exploration). Solving it well is necessary for long-run reward maximisation.

Credit Assignment. The problem, central to reinforcement learning and to deep network training, of determining which earlier decisions or parameters are responsible for an eventual outcome. Long-horizon credit assignment remains a frontier problem.

Classification, Regression, Generation. The three canonical task types: classification predicts a discrete label, regression a continuous value, and generation a sample from a high-dimensional output space. Each is paired with a characteristic loss function.

Sequence Modelling. The task of learning probability distributions over ordered sequences, typically by factorising $p(x_1 \ldots x\_T)$ into a product of next-token conditionals. The mathematical foundation of language modelling.

Representation

How AI builds abstractions

Modern AI succeeds because it learns its own representations. The Representation lens treats those representations as geometric objects — points, manifolds, latent spaces — and connects compression, transfer and interpretability through a unified geometric view.

Representation Learning. The automatic discovery of transformations $\varphi_\theta : X \to Z$ that make downstream tasks easier — linearly separable, smoothly varying, disentangled, or compressed in the information-bottleneck sense. The defining shift from hand-crafted features to learned ones.

Feature vs Representation. A feature is a single scalar measurement extracted from the input; a representation is a coordinated vector of features jointly optimised to encode the structure of the data. Distributed representations are exponentially more expressive because meaning resides in the pattern, not in any single dimension.

Distributed Representation. A representation in which each concept is encoded by a pattern of activity across many units, and each unit participates in encoding many concepts. The standard form of representation in modern neural networks.

Embedding. A learned vector representation of a discrete object — a word, a token, an entity — situated in a continuous space so that semantic relationships correspond to geometric ones.

Latent Space. The internal vector space in which a model's representations live. Treated geometrically, it is a Riemannian manifold whose distances, geodesics and curvature encode learned structure.

Manifold Hypothesis. The proposition that natural data — images, language, audio, protein structures — does not fill high-dimensional space uniformly but lies on or near a low-dimensional manifold within it. The effective dimensionality of the learning problem is governed by the manifold's intrinsic dimension, not the ambient one.

Intrinsic Dimension. The true dimensionality of the data manifold, typically far smaller than the ambient dimension of the input space. Learning algorithms succeed in part because they discover and operate on this intrinsic structure.

Information Bottleneck. Tishby, Pereira and Bialek's criterion that a good representation maximises mutual information with the target Y while minimising mutual information with the input X. A formal account of what a representation should keep and what it should discard.

Compression as Intelligence. The view, traceable to Solomonoff, that a model which describes data with a short program has discovered genuine structure. Memorisation is not compression; compression is a proxy for understanding.

Distributional Hypothesis. The linguistic principle that words occurring in similar contexts tend to have similar meanings. Operationalised by Word2Vec, GloVe and the contextual embeddings that followed.

Contextual Embeddings. Token representations whose values depend on the surrounding context, in contrast to fixed type-level embeddings. Introduced by ELMo and made standard by BERT and GPT.

Disentanglement. The property that distinct factors of variation in the data are encoded along distinct directions of the representation, so that changing one factor leaves the others unchanged.

Linear Separability. The property of a representation that allows classes to be separated by a hyperplane. Linear separability in the final layer is a common diagnostic for whether a representation has captured the relevant structure.

Riemannian Manifold (in ML). A latent space treated as a smooth manifold equipped with a metric, so that notions of length, angle, geodesic and curvature can be applied. The right object for reasoning about interpolation, traversal and the geometry of learned representations.

Spherical Interpolation (Slerp). Interpolation between two points along a great-circle arc on a unit sphere, used as an empirical probe of latent-space structure. Coherent slerp trajectories are evidence that the latent geometry is meaningful.

Hidden Function. The function that a trained network implicitly computes, encoded across billions of parameters and not directly readable from inspection. Best understood as a computational procedure rather than a closed-form expression.

Neural Networks as Programs. The view that a parameter vector, together with an architecture, specifies a computational procedure that solves a task. AlphaFold is the canonical example: a procedure too complex to engineer manually but discoverable through learning.

Multimodal Alignment. The phenomenon, prototyped by CLIP, in which models trained on different modalities (image, text, audio) converge toward a shared geometric structure in their representation spaces.

Platonic Representation Hypothesis. The conjecture that sufficiently large models, trained on sufficiently rich data from different modalities, converge toward a common underlying representation of the world.

Language as a Manifold. The view that coherent language occupies a thin manifold within the astronomically large combinatorial space of token sequences, sculpted by syntactic, semantic, discourse and world-knowledge constraints. Explains why next-token prediction is so effective and why scaling helps.

Computation

How intelligence is implemented

Once the problem is formulated and the representation strategy chosen, computation is the actual machinery that performs the work. The Computation lens exposes the small set of primitives — affine maps, nonlinearities, convolution, attention, residual connections — from which any modern architecture is built.

Neural Network. A parameterised function composed of layers, each performing an affine transformation followed by a nonlinear activation. Without nonlinearity, stacking layers collapses to a single linear map.

Architecture. The pre-training design specification — depth, width, layer types, connectivity, parameter count — that defines the hypothesis space and encodes assumptions about the structure of the problem.

Multi-Layer Perceptron (MLP). A fully connected feedforward network with no structural assumptions about its input. Universal in principle but data- and compute-hungry in practice; survives mainly as a component of larger architectures.

Convolutional Neural Network (CNN). An architecture designed for data with spatial structure: small filters slide across the input, the same filter applied at every position. Encodes the inductive biases of locality and translation equivariance.

Transformer. An architecture in which attention dynamically routes information between every pair of positions in the input. Inductive bias is relational rather than spatial or sequential; cost is quadratic in sequence length.

Attention Mechanism. A differentiable operation in which queries, keys and values — each a linear projection of the input — combine through a softmax over compatibility scores to produce weighted averages of values. Reads as a learnable kernel regression or as a soft, content-addressable lookup.

Query, Key, Value. The three projections of the input used by scaled dot-product attention. Queries select what to look for, keys advertise what each position offers, and values carry the content that is ultimately mixed.

Multi-Head Attention. A partition of the representation into h parallel subspaces, each with its own attention computation. Different heads specialise — empirically — for positional, syntactic and semantic relationships without any explicit supervision.

Positional Encoding. Information about token position injected into a transformer's input, because attention is otherwise permutation-equivariant. Schemes include sinusoidal, learned, rotary (RoPE) and ALiBi, each with distinct mathematical properties.

Rotary Positional Embedding (RoPE). A positional scheme that applies position multiplicatively as a rotation, so that the dot product between two embeddings depends only on the relative offset between them.

Residual Connection. A connection that adds the input of a sublayer to its output, so that gradients flow directly through depth. Enables stable training of very deep networks.

Residual Stream. The depth-wise interpretation of the transformer in which the representation at any layer is the sum of the original embedding and all updates applied by preceding layers. The residual stream functions as a broadcast communication channel across the network.

Feed-Forward Network (in Transformers). The per-position two-layer MLP that follows attention in each transformer block. Increasingly understood as an associative memory whose first layer encodes keys and second layer encodes values.

Layer Normalisation. A normalisation applied across features within each token, stabilising the distribution of activations. Pre-norm (before the sublayer) is now standard; post-norm preceded it and remains in some architectures.

Activation Function. The nonlinearity applied elementwise after each affine transformation. ReLU, GELU and SwiGLU are the standard choices in modern networks; without nonlinearity, no expressive gain is achieved by stacking layers.

Universal Approximation. The classical result that sufficiently wide (or sufficiently deep) feedforward networks can approximate any continuous function on a compact domain to arbitrary accuracy. A necessary but insufficient explanation of deep learning's success.

Translation Equivariance / Invariance. The property that a transformation applied to the input produces a correspondingly transformed (equivariant) or unchanged (invariant) output. The inductive bias that makes convolution so effective on image data.

Permutation Equivariance. The property that the output of a model is reordered consistently when the input is reordered. Attention is permutation-equivariant, which is why positional encoding is required.

Vision Transformer (ViT). A transformer applied to images by treating an image as a sequence of patches. Matches or exceeds convolutional architectures despite using none of their image-specific inductive biases.

FlashAttention. An efficient implementation of exact attention that reduces memory traffic by tiling the computation and recomputing intermediate values during the backward pass. Representative of the systems-level work that has made long-context transformers practical.

Optimisation

Discovering useful solutions

Architecture defines a hypothesis space; loss defines what success means; optimisation is the search by which good parameters are actually found. The Optimisation lens covers gradient descent and its modern variants, the geometry of high-dimensional loss landscapes, the family of regularisation methods, and the implicit biases that make non-convex optimisation work in practice.

Loss Function. A non-negative scalar map measuring the discrepancy between predictions and targets. The choice of loss is the mathematical specification of what learning means; MSE, cross-entropy and hinge are the canonical examples.

Loss Landscape. The loss viewed as a surface over parameter space. In high dimensions it is qualitatively unlike low-dimensional landscapes — dominated by saddle points, with vast connected regions of near-equal loss.

Gradient Descent. The canonical first-order optimisation algorithm: compute the gradient of the loss, take a step in the opposite direction, repeat. The learning rate controls step size and is the single most consequential hyperparameter.

Stochastic Gradient Descent (SGD). Gradient descent in which each update uses the gradient on a randomly sampled mini-batch rather than the full dataset. Cheap, noisy, and implicitly regularising; the workhorse of modern deep learning.

Mini-Batch. A small randomly drawn subset of the training set used to estimate the gradient at each optimisation step. The size trades off compute, gradient noise and statistical accuracy.

Momentum. An exponentially weighted moving average of past gradients, added to the current step. Accelerates descent through long shallow valleys and damps oscillations across narrow ones.

Adam / AdamW. Adaptive optimisers that combine first-moment (momentum) and second-moment (RMSProp-style) estimates of the gradient. AdamW decouples weight decay from the adaptive update and is the modern default.

Learning Rate Schedule. A function specifying how the learning rate changes over training. Warmup followed by cosine annealing is the de-facto default for large language model pre-training.

Backpropagation. The efficient algorithm for computing gradients of a composite function by applying the chain rule layer by layer in reverse. Its contribution is computational, not conceptual: it makes the otherwise intractable gradient computation cheap.

Regularisation. A modification to the optimisation objective intended to reduce overfitting, typically by encoding a prior belief about what plausible solutions look like. The family includes L1, L2, elastic net, dropout, weight decay and data augmentation.

Ridge Regression (L2). Linear regression with an L2 penalty on the coefficients, yielding the closed-form estimator $(X^TX + \lambda I)^{-1}X^Ty$. Geometrically, it constrains the solution to an L2 ball whose radius shrinks with λ.

Lasso (L1). Linear regression with an L1 penalty. The diamond geometry of the constraint region produces sparse solutions, so Lasso performs automatic feature selection alongside estimation.

Elastic Net. A mixed L1+L2 penalty that combines Lasso's sparsity with Ridge's stability under correlated predictors. A mixing parameter interpolates between the two extremes.

Bias–Variance Trade-off. The decomposition of expected squared error into bias and variance components, controlled in regularised regression by the penalty strength λ. The classical U-curve in test error is its visual signature.

Bayesian Interpretation of Regularisation. The reading of regularised estimation as maximum-a-posteriori inference under a specific prior — Gaussian for ridge, Laplace for Lasso, mixed for elastic net. Choosing a penalty is choosing a prior.

Maximum Likelihood Estimation (MLE). The choice of parameters that maximise the probability of the observed data under the model. Minimising MSE is MLE under Gaussian noise; minimising cross-entropy is MLE under a categorical model.

Maximum-a-Posteriori (MAP) Estimation. The choice of parameters that maximise their posterior probability given the data. Equivalent to penalised MLE, where the penalty is the negative log of the prior.

Cross-Entropy Loss. The standard loss for classification: the inefficiency, measured in nats or bits, of encoding samples from the true distribution under the model's predicted distribution. Minimising cross-entropy is exactly minimising the KL divergence to the empirical distribution.

Flat Minima. Regions of the loss landscape where the loss varies slowly with the parameters. Empirically associated with better generalisation; SGD's gradient noise is believed to bias trajectories toward them.

Implicit Regularisation. Generalisation-improving structure not explicitly added to the loss, arising instead from architecture, the inductive bias of gradient descent, the noise of stochastic optimisation, and initialisation. Helps explain why modern over-parameterised networks generalise without aggressive explicit regularisation.

Theory

Why AI works

Classical statistical learning theory predicted that modern over-parameterised networks should fail, yet they succeed spectacularly. The Theory lens surveys the most useful partial explanations now available — implicit regularisation, the manifold hypothesis, double descent, benign overfitting, scaling laws, mechanistic interpretability — and is honest that the destination is a frontier rather than a closed proof.

Classical Statistical Learning Theory. The framework — bias–variance, VC dimension, Rademacher complexity, uniform convergence, structural risk minimisation — that dominated statistical learning for half a century. Mathematically correct within its assumptions but developed for a regime quite different from modern gradient-trained deep networks.

VC Dimension. The Vapnik–Chervonenkis dimension: the largest set of points a hypothesis class can shatter (label in all possible ways). The classical measure of capacity, which predicts catastrophic overfitting for modern over-parameterised networks — and so requires augmentation in the deep-learning regime.

Rademacher Complexity. A data-dependent measure of a hypothesis class's ability to fit random noise. Tighter than VC dimension in many cases, but, like VC dimension, predicts behaviour that does not match modern empirical results.

Uniform Convergence. The classical guarantee that the empirical risk converges to the true risk uniformly over a hypothesis class. The foundation of generalisation bounds in statistical learning theory, but inadequate as a complete account of modern deep networks.

Overparameterisation. The regime in which the number of parameters p vastly exceeds the number of training examples n. Classical theory predicts catastrophic overfitting; modern practice shows generalisation that improves with scale.

Interpolation Threshold. The model capacity at which the training error first reaches zero. Classical theory says generalisation should be worst here; the empirical curve in fact spikes at the threshold and then improves as capacity grows further.

Double Descent. Belkin and colleagues' observation that, as model complexity is pushed beyond the interpolation threshold, test error decreases again — sometimes below the classical optimum. The full curve has three regions: classical, interpolation spike, and over-parameterised.

Benign Overfitting. The phenomenon, made precise by Bartlett, Tsigler and others, that modern networks can perfectly fit noisy training data and still generalise. Connected to the manifold hypothesis and to the implicit bias of gradient descent toward minimum-norm interpolators.

Implicit Bias of Gradient Descent. The tendency of gradient-based optimisation, even without explicit regularisation, to select solutions with particular structural properties —

minimum norm, maximum margin, low-rank. A central ingredient in modern explanations of generalisation.

Minimum-Norm Interpolating Solution. Among all parameter settings that perfectly fit the training data, the one with the smallest norm. Gradient descent from small initialisation is observed (and in linear cases proved) to converge to such solutions, which generalise far better than arbitrary interpolators.

Scaling Laws. The empirical observation, formalised by Kaplan and colleagues, that cross-entropy loss decreases as a clean power law in model parameters, training tokens and compute budget. Scaling laws predict loss, not capabilities.

Chinchilla Law. Hoffmann and colleagues' refinement of the Kaplan scaling laws, showing that compute-optimal training splits resources roughly equally between model size and dataset size. Retroactively reframed pre-2022 LLMs as significantly under-trained.

Emergent Abilities. Capabilities — multi-step arithmetic, chain-of-thought reasoning, in-context learning — that appear abruptly at specific model scales rather than improving smoothly. Catalogued by Wei and colleagues; scaling laws predict loss but are explicitly silent on which capabilities a given loss level will unlock.

Lottery Ticket Hypothesis. Frankle and Carbin's claim that a randomly initialised dense network contains a sparse subnetwork — the winning ticket — which, when trained in isolation from its original initialisation, reaches comparable accuracy. Recasts training as the identification of an existing capable subnetwork within the over-parameterised hypothesis space.

Neural Tangent Kernel (NTK). The kernel obtained by linearising a wide neural network around its initialisation. Provides a tractable theoretical regime in which infinitely wide networks behave like kernel methods and gradient descent admits closed-form analysis.

PAC-Bayes. A family of generalisation bounds that hold over posterior distributions on parameters, often delivering non-vacuous guarantees for neural networks where uniform-convergence bounds collapse.

Entropy. Shannon's measure of average surprise in a distribution. The theoretical foundation under which cross-entropy loss is read as a negative log-likelihood and under which compression and learning become formally connected.

KL Divergence. A non-symmetric measure of difference between two distributions, equal to the expected log-ratio. Minimising cross-entropy in training is exactly minimising KL from the model to the empirical distribution.

Mutual Information. The reduction in uncertainty about one random variable given knowledge of another. The currency in which the information bottleneck principle and information-theoretic generalisation bounds are stated.

Minimum Description Length (MDL). Rissanen's principle that the best model is the one that gives the shortest combined description of the model and the data given the model. A formal expression of the link between compression, learning and generalisation.

Frontiers

Current research and open questions

Where the framework continues to apply but the answers are still being worked out. The Frontiers lens surveys the major active research programmes — interpretability, scaling, world models, scientific machine learning, neuro-symbolic systems, agentic AI — and the open questions that animate each.

Mechanistic Interpretability. The research programme of reverse-engineering neural networks at the level of internal computations — not just what they do but how they do it. Treated as theory rather than tooling, since circuit-level findings constrain and inform any account of why networks generalise.

Circuits. Small, identifiable subnetworks within a trained model that implement specific computational functions. The unit of analysis in mechanistic interpretability.

Induction Heads. Attention heads, discovered by Olsson and colleagues, that implement a copy-and-continue pattern (look back to a previous occurrence, attend to what followed, emit that token). Believed to underlie in-context learning.

Indirect Object Identification (IOI). A canonical case study in mechanistic interpretability identifying the circuit by which a transformer determines the indirect object of a sentence. Exemplifies the move from individual neuron analysis to circuit-level theory.

Superposition. The phenomenon in which a network represents more features than it has neurons, by storing them as near-orthogonal directions in the activation space. A central obstacle to clean interpretability and an active research target.

World Models. Learned models of environment dynamics that support planning, simulation and counterfactual reasoning. The integration of representation learning with structured prediction; central to reinforcement learning, robotics and embodied AI.

Scientific Machine Learning. The application of machine learning to scientific discovery: physics-informed neural networks, neural differential equations, learned surrogates for expensive simulations, and the discovery of new materials, molecules and proteins. AlphaFold is the exemplar.

Physics-Informed Neural Networks (PINNs). Networks whose loss includes a penalty for violating known physical laws, typically expressed as residuals of differential equations. A representative technique for combining data-driven learning with first-principles knowledge.

Neural Differential Equations. Architectures in which the depth of the network becomes the continuous time variable of an ODE, parameterised by a learned vector field. A bridge between deep learning and classical dynamical systems.

Symbolic AI. The first four decades of AI, dominated by logical propositions, production rules, semantic networks, frames and ontologies. The contrast against which modern statistical approaches are defined.

Neuro-Symbolic Systems. Architectures that combine the pattern recognition of neural networks with the reasoning and verifiability of symbolic systems. A live research direction motivated by the limitations of pure statistical models on tasks requiring composition and proof.

Foundation Models. The logical endpoint of transfer learning: a single, broadly-trained representation that amortises the cost of learning general structure across all downstream tasks. Defined by pre-training objective, scale and adaptability rather than by a particular architecture.

Multimodal AI. Systems that process inputs from multiple modalities — vision, audio, language — in a shared representation space, usually by tokenising each modality and feeding the union into a single transformer.

Agentic AI. Systems in which a foundation model is integrated with tools, memory and an environment, so that it plans, acts, observes outcomes and replans. Raises open research questions in long-horizon credit assignment, tool use, multi-agent coordination and alignment.

Tool Use. The capability of a model to invoke external functions — calculators, search engines, code interpreters, APIs — and incorporate their outputs into its reasoning. A key mechanism by which agentic systems extend beyond their parametric knowledge.

Tree Ensembles. Gradient-boosted decision tree ensembles (XGBoost, LightGBM, CatBoost) which continue to dominate on tabular data despite the rise of deep learning. Tree splits handle semantic heterogeneity natively; neural networks must learn it.

Kernel Methods. A family of non-neural learners — including SVMs and kernel ridge regression — that work in implicit high-dimensional feature spaces via the kernel trick. Retain theoretical guarantees that deep networks have not yet matched.

Gaussian Processes. Distributions over functions used as priors in Bayesian non-parametric regression. Provide calibrated uncertainty estimates that are difficult to obtain from standard deep networks.

Diffusion Models. A class of generative model that gradually destroys data with Gaussian noise in a forward process and learns to reverse the noising step by step. Training uses a simple MSE objective on predicted noise; sampling is iterative.

Open Research Questions. The animating questions of the present moment: why deep learning generalises, what the correct complexity measure for neural networks is, whether emergent capabilities can be predicted, how to align increasingly capable systems with human intent, and what comes after the transformer.